

NEURO

The Living Digital Mascot

A product by NeuroGrid Labs

Bringing ideas into life.

What is Neuro?

Neuro is a web-based AI avatar — a glowing biometric-scan face that sees you, listens, thinks, and speaks back. It runs in any modern web browser, no install required. The same Neuro can be embedded on a website, on a tablet at a reception desk, in a kiosk, or behind a WhatsApp number.

Think of it as a receptionist, a salesperson, a teaching assistant, and a 24/7 brand ambassador — combined into a single deployable product. One engine. Infinite faces. Fully white-labelled per client.

What it does

Capability	What the visitor experiences
Visual presence	A 3D cyan biometric-scan face fills the screen. It blinks, breathes, and tracks the person through the webcam — without ever showing or recording the visitor's image.
Sleeps & wakes	Eyes close when no one is in front of the camera. They open the moment a face is detected, the screen is touched, or someone speaks.
Eyes that follow	Gold camera-style focus reticles track the visitor's position in real time. Head turns subtly to face them.
Listens & speaks	Voice in (microphone), voice out (speakers). Hands-free conversation. Live captions appear below for accessibility.
Touch-reactive	Tapping the avatar's face triggers a positive flash and ripple — it "feels" the contact.
Powered by Claude	Anthropic's most capable language model interprets the conversation and decides what to say, when to look up information, and how to capture lead details.
Live knowledge base	Reads pricing, product info, and FAQ data from a Google Sheet the client controls. Edit the sheet !' Neuro speaks the new info on the next question. No code, no redeploy.
Lead capture pipeline	When a visitor shares their name, email, phone, or interest, Neuro writes a row to the Leads tab in the same sheet, in real time. The team sees leads land instantly.
Multi-channel brain	The same engine powers the web avatar AND a WhatsApp Business number. A customer asks on WhatsApp !' Neuro replies with the same personality, same knowledge, same lead capture.
White-labelled	One deployment, infinite brands. A URL like ?brand=falconhouse switches greeting, personality, brand colors, and voice — perfect for schools, clinics, retail chains, hotels.

Where it can be deployed

- **Schools** — Front-desk attendant that answers parent questions about admissions, fees, and timings.
- **Clinics & hospitals** — Patient triage and appointment booking, in English or Urdu.
- **Retail & e-commerce** — Brand mascot on the website plus WhatsApp; turns browsers into qualified leads.
- **Events & expos** — Interactive booth attendant that captures visitor info while pitching products.
- **Government / civic** — Public-facing information point for citizens, in multiple languages.
- **Hotels & hospitality** — 24/7 concierge that handles bookings, restaurant info, and check-in queries.
- **Corporate reception** — Visitor greeting + intent routing, freeing up human receptionists for complex requests.

Technology stack

Application layer

Layer	Technology
Frontend framework	Next.js 16 (App Router) + React 19 + TypeScript
Styling	Tailwind CSS 4
3D rendering	Three.js + React Three Fiber + @react-three/drei
Animation	Custom JS animation loops with smooth interpolation
Hosting	Vercel (serverless) — or any Node-compatible host

AI & vision

Layer	Technology
Face tracking	MediaPipe FaceLandmarker (Google) — 468 facial landmarks, runs in-browser
Voice input	Web Speech API (SpeechRecognition) — live transcription with interim results
Voice output	Web Speech API (SpeechSynthesis) — customizable voice + rate + pitch
AI brain	Anthropic Claude API (claude-sonnet-4-5) with native tool use + prompt caching

Integrations

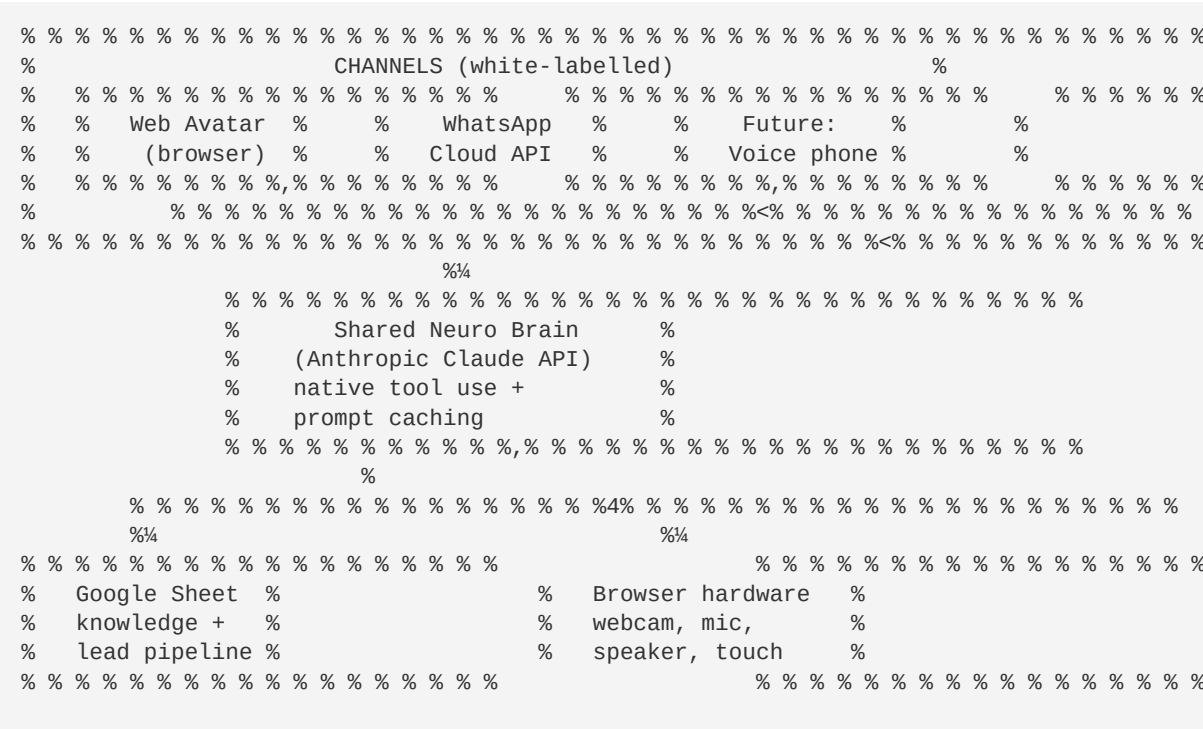
Layer	Technology
Knowledge base (read)	Google Sheets API v4 via service-account authentication
Lead pipeline (write)	Google Sheets API v4 (append rows to a Leads tab)
WhatsApp channel	Meta Graph API v22.0 (WhatsApp Cloud API) — webhook receive + send
Per-user conversation	In-memory store (production-ready upgrade path: Redis / PostgreSQL)

Design system

- **Brand palette** — Dark #080c0e · Emerald #10b981 · Gold #d4af37
- **Avatar palette** — Biometric cyan #3ad5ff with pure-white iris cores (additive blending !' glows like a screen)

- **Typography** — Geist Sans + Geist Mono (Vercel typeface)
- **Aesthetic** — Biometric scan portrait — point cloud + wireframe + glowing camera-reticle eyes

Architecture



The shared brain pattern means: a change to Neuro's personality, knowledge, or behavior automatically propagates to every channel at once. Update the Google Sheet !' web visitors and WhatsApp users instantly hear the new prices.

Current build status

Phases shipped (production-quality)

1. Fullscreen shell + reactive neural-particle background
2. Biometric-scan 3D face (real point-cloud topology + wireframe + reticle eyes)
3. Real-time webcam face tracking via MediaPipe (no video stored)
4. Sleep / wake on idle + detection + touch
5. Touch reaction (positive flash + iris pulse)
6. Voice loop (mic !' Claude !' speaker) with live captions
7. Action markers (open URL, capture lead, show product) + brand white-labelling
8. Google Sheets knowledge base with Anthropic native tool-use
9. Lead pipeline !' live Google Sheet rows in real time
10. WhatsApp Cloud API connector (Meta integration)

On the roadmap

- Wake-word detection ("Hey Neuro") via Picovoice Porcupine
- Persistent cross-session memory (returning users greeted by name)
- Branded TTS voice via ElevenLabs (replacing browser default)
- Refined head topology using MediaPipe canonical face mesh
- Multi-language support — Urdu, Sindhi, Punjabi, Arabic
- Direct CRM hooks (HubSpot, Salesforce) alongside the Sheets pipeline

About NeuroGrid Labs

NeuroGrid Labs is a Karachi-based deep-tech company founded **April 5, 2026** by **Sheikh Shumyle Shafiq** — INFJ visionary, BS Computer Science (Karachi University), British Council APTIS English fluent, EC-Council Cybersecurity certified.

The company builds across six verticals — EdTech, Healthcare, Mobility, Civic Tech, Security, and AI Agents — with 33+ products in active development or live deployment.

Live deployments

- **SchoolOS** — Complete student management system — schoolportal.live
- **NeuroGrid Clinic** — Healthcare practice management — karachitraumainstitute.netlify.app
- **CivicPulse** — Public-issue reporting and accountability — civicpulsepk.netlify.app

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